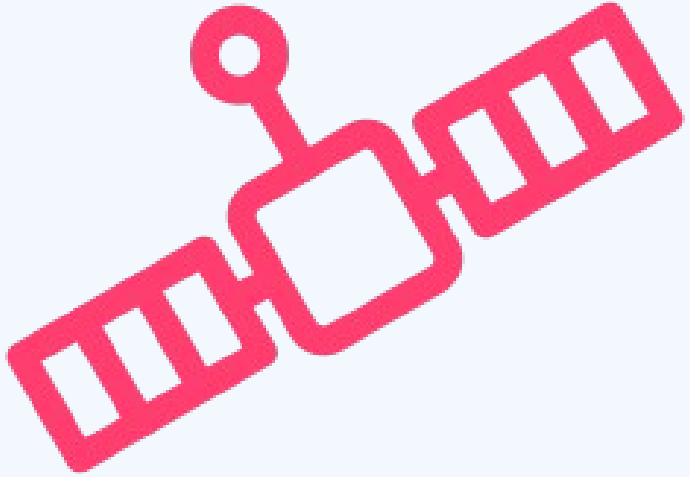
SOFTWARE SYSTEMS / PROJECT CONCEPT

Satellite Predictive Maintenance

Estimating reaction-wheel remaining useful life from sensor history.



Predictive maintenance works across domains.

Use machine measurements and operational context to anticipate degradation instead of reacting after a breakdown.

Industrial machinery

Motors, gearboxes, factory bearings

Wind turbines

Rotating shafts, generator bearings

Satellite systems

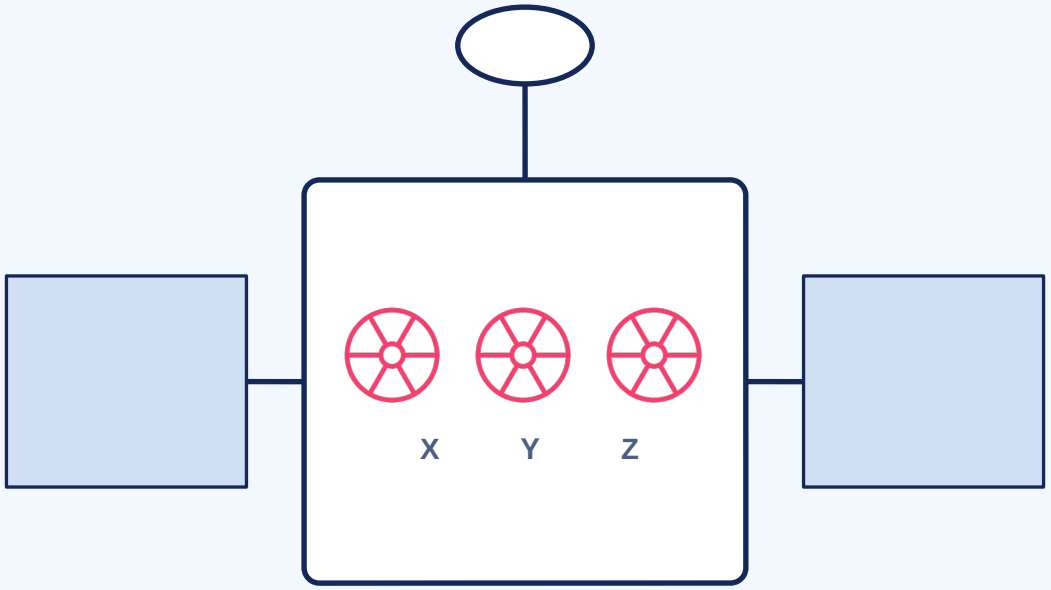
Reaction-wheel mechanisms

The general idea

Sensor data → **Software** → **Maintenance insight**

What is a reaction wheel?

A spinning wheel inside the satellite that keeps it pointed in the right direction.



Reaction wheels, inside the satellite body

Where

Inside the satellite body, mounted along different axes: usually three wheels or more.

What it does

Speeding the wheel up or slowing it down turns the satellite the opposite way, without using fuel.

What it is for

Keeps antennas, cameras and solar panels pointed where they need to be.

Why we monitor it: moving parts that run for years and cannot be repaired in orbit.

NASA Kepler: an operational warning.

A documented case where reaction-wheel deterioration changed a space mission.

JUL 2012

First wheel failed

Kepler lost reaction wheel 2.

EARLY 2013

Friction symptoms

Increasing friction appeared in wheel 4; engineers attempted life-extension measures.

MAY 2013

Wheel 4 failed

Only two working wheels remained.



3 of 4

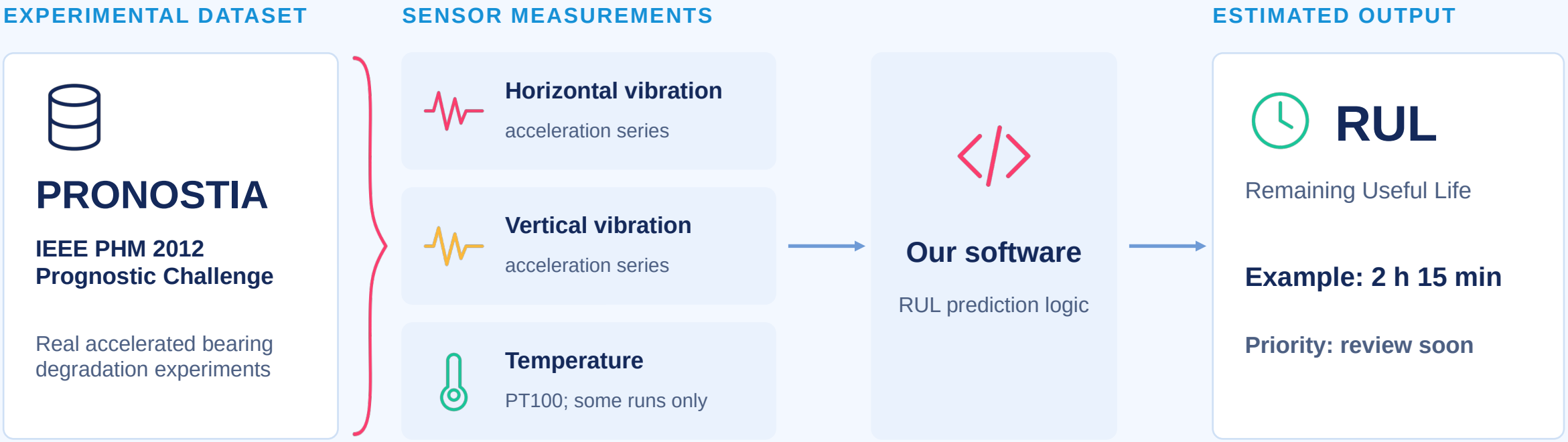
wheels required for original precision pointing

Why forecast? More lead time to review wheel health, adapt operations and prepare contingencies.

NASA (2013) and NESC/Dennehy (2014). Warnings do not guarantee prevention of failure.

From Sensor Measurements to Remaining Useful Life

PRONOSTIA bearing measurements enter our software; the output is an estimate of remaining operating time.



Nectoux et al. (IEEE PHM 2012), “PRONOSTIA: An Experimental Platform...” • Bearing benchmark, not satellite flight data.

AI agents support our development.

Focused coding assistance throughout implementation, testing and integration.

01



Data & features

Implement dataset readers, validation and sensor-feature extraction.

02



RUL component

Translate our chosen estimation approach into modular, testable code.

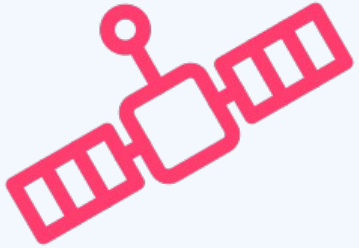
03



Tests & integration

Generate tests, investigate bugs, refactor and integrate team changes.

Our team defines the requirements, reviews the code and owns the final result.



Thank you for your attention

Satellite Predictive Maintenance